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When the blood supply to the heart muscle or myocardium is remarkably reduced it leads to death of the muscle fibres. This condition is called heart attack or myocardial infarction. The blood clot or thrombosis blocks the blood supply to the heart and weakens the muscle fibres. It is also called **Ischemic** heart disease due to lack of oxygen supply to the heart muscles. If this persists it leads to chest pain or angina. Prolonged angina leads to death of the heart muscle resulting in heart failure.

Rheumatoid Heart Disease

Rheumatic fever is an autoimmune disease which occurs 2-4 weeks after throat infection usually a streptococcal infection. The antibodies developed to combat the infection cause damage to the heart. Effects include fibrous nodules on the mitral valve, fibrosis of the connective tissue and accumulation of fluid in the pericardial cavity.

Varicose veins The veins are so dilated that the valves prevent back flow of blood. The veins lose their elasticity and become congested. Common sites are legs, rectal-anal regions (haemorrhoids), the oesophagus and the spermatic cord.

Embolism is the obstruction of the blood vessel by abnormal mass of materials such as fragment of the blood clot, bone fragment or an air bubble. Embolus may lodge in the lungs, coronary artery or liver and leads to death.

Aneurysm The weakened regions of the wall of the artery or veins bulges to form a balloon like sac. Unruptured aneurysm may exert pressure on the adjacent tissues or may burst causing massive haemorrhage.

7.8 Diagnosis and Treatment

Angiogram

Angiogram is a procedure that uses a special dye and X-ray to see how blood flows through the coronary arteries of the heart and it can be used to detect abnormality in the blood vessels through out the body.

Angioplasty

Angioplasty is the stretching of an artery that is narrowed due to atherosclerosis. The risk involved in this procedure is minimal. During an angioplasty a small long balloon catheter is threaded through the blocked artery. A deflated balloon is attached to the catheter and the balloon is inflated to widen the arterial wall. Then the tube and the balloon are removed. A small metal scaffold called stent is left in place. This scaffolding keeps the blood vessel open and allows free flow of blood. Slow releasing stents are now available that can release chemicals to prevent further block of the artery.

Bypass Surgery

When the arteries that bring blood to the heart muscles (coronary artery) are blocked by plaque (accumulation of fat, cholesterol and other substances) the person is advised to undergo Bypass surgery. After the surgery the blood flow to coronary artery is increased and the person is relieved from chest pain. This is a major surgery where damaged blood vessel is replaced by the healthy one taken from